

hovid-Duloxetine Capsule 30 mg

VIDULxx-PC

DESCRIPTION

Dark blue opaque blue and white opaque capsule.

COMPOSITION

Each capsule contains: Duloxetine pellets equivalent to Duloxetine 30 mg

ACTIONS AND PHARMACOLOGY

Duloxetine is a combined serotonin (5-HT) and noradrenaline (NA) reuptake inhibitor. It weakly inhibits dopamine reuptake, with no significant affinity for histaminergic, dopaminergic, cholinergic, and adrenergic receptors. Duloxetine dose-dependently increases extracellular levels of serotonin and noradrenaline in various brain areas of animals.

PHARMACOKINETICS

Absorption: Duloxetine is well absorbed after oral administration, with a C_{max} occurring 6 hours post-dose. The absolute oral bioavailability of Duloxetine ranged from 32% to 80% (mean of 50%). Food delays the time to reach the peak concentration from 6 to 10 hours and it marginally decreases the extent of absorption (approximately 11%). These changes do not have any clinical significance.

Distribution: Duloxetine is approximately 96% bound to human plasma proteins. Duloxetine binds to both albumin and alpha1-acid glycoprotein. Protein binding is not affected by renal or hepatic impairment.

Metabolism: Duloxetine is extensively metabolised and the metabolites are excreted principally in urine. Both cytochromes P450-2D6 and 1A2 catalyse the formation of the two major metabolites, glucuronide conjugate of 4-hydroxy Duloxetine and sulphate conjugate of 5-hydroxy, 6-methoxy Duloxetine. Based upon in vitro studies, the circulating metabolites of Duloxetine are considered pharmacologically inactive. The pharmacokinetics of Duloxetine in patients who are poor metabolisers with respect to CYP2D6 has not been specifically investigated. Limited data suggest that the plasma levels of Duloxetine are higher in these patients.

Elimination: The elimination half-life of Duloxetine ranges from 8 to 17 hours (mean of 12 hours). After an oral dose the apparent plasma clearance of Duloxetine ranges from 33 to 261 l/hr (mean 101 l/hr).

INDICATIONS

- Treatment of major depressive disorder,
- Management of neuropathic pain associated with diabetic peripheral neuropathy in adults
- Treatment of generalized anxiety disorder

CONTRAINDICATIONS

Duloxetine is contraindicated in patients with hypersensitivity to the active substance or any of the excipients.

Concomitant use of Duloxetine with non-selective, irreversible monoamine oxidase inhibitors (MAOIs) is contraindicated.

Liver disease resulting in hepatic impairment.

Duloxetine should not be used in combination with fluvoxamine, ciprofloxacin or enoxacin (i.e., potent CYP1A2 inhibitors), since the combination results in elevated plasma concentrations of Duloxetine.

Increased plasma concentrations of Duloxetine occur in patients with severe renal impairment on haemodialysis (creatinine clearance <30 ml/min). Duloxetine must not be used in patients with severe renal impairment (creatinine clearance <30 ml/min).

The initiation of treatment with Duloxetine is contraindicated in patients with uncontrolled hypertension that could expose patients to a potential risk of hypertensive crisis.

WARNINGS AND PRECAUTIONS

Mania and Seizures: Duloxetine should be used with caution in patients with a history of mania or a diagnosis of bipolar disorder, and/or seizures.

Mydriasis: Mydriasis has been reported in association with Duloxetine, therefore, caution should be used when prescribing Duloxetine to patients with increased intraocular pressure or those at risk of acute narrow-angle glaucoma.

Blood Pressure and Heart Rate: Duloxetine has been associated with an increase in blood pressure, and clinically significant hypertension in some patients. This may be due to the noradrenergic effect of Duloxetine.

Cases of hypertensive crisis have been reported with Duloxetine, especially in patients with pre-existing hypertension. Therefore, in patients with known hypertension and/or other cardiac disease, blood pressure monitoring is recommended, especially during the first month of treatment. Duloxetine should be used with caution in patients whose conditions could be compromised by an increased heart rate or by an increase in blood pressure. Caution should also be exercised when Duloxetine is used with medicinal products that may impair its metabolism. For patients who experience a sustained increase in blood pressure while receiving Duloxetine, either dose reduction or gradual discontinuation should be considered. In patients with uncontrolled hypertension, Duloxetine should not be initiated

Serotonin syndrome: As with other serotonergic agents, serotonin syndrome, a potentially life-threatening condition, may occur with Duloxetine treatment, particularly with concomitant use of other serotonergic agents (including SSRIs, SNRIs tricyclic antidepressants or triptans), with agents that impair metabolism of serotonin such as MAOIs, or with antipsychotics or other dopamine antagonists that may affect the serotonergic neurotransmitter systems.

Serotonin syndrome symptoms may include mental status changes (e.g., agitation, hallucinations, coma), autonomic instability (e.g., tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g., hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g., nausea, vomiting, diarrhoea).

If concomitant treatment with Duloxetine and other serotonergic agents that may affect the serotonergic and/or dopaminergic neurotransmitter systems is clinically warranted, careful observation of the patient is advised, particularly during treatment initiation and dose increases.

St John's wort: Adverse reactions may be more common during concomitant use of Duloxetine and herbal preparations containing St John's Wort (*Hypericum perforatum*).

Suicide: Cases of suicidal thoughts and suicidal behaviours have been reported during duloxetine therapy or early after treatment discontinuation in patients treated for major depressive disorder, diabetic peripheral neuropathic pain, children and adolescents under 18 years. Other psychiatric conditions for which Duloxetine is prescribed can also be associated with an increased risk of suicide-related events. Patients with a history of suicide-related events or those exhibiting a significant degree of suicidal thoughts prior to commencement of treatment, are known to be at greater risk of suicidal thoughts or suicidal behaviour, and should receive careful monitoring during treatment. A meta-analysis of placebo-controlled clinical trials of antidepressant medicinal products in psychiatric disorders showed an increased risk of suicidal behaviour with antidepressants compared to placebo in patients less than 25 years old.

Close supervision of patients, and in particular those at high risk, should accompany medicinal product therapy, especially in early treatment and following dose changes. Patients (and caregivers of patients) should be alerted about the need to monitor for any clinical worsening, suicidal behaviour or thoughts, and unusual changes in behaviour, and to seek medical advice immediately if these symptoms present.

Haemorrhage: There have been reports of bleeding abnormalities, such as ecchymoses, purpura, and gastrointestinal haemorrhage, with selective serotonin reuptake inhibitors (SSRIs) and serotonin/noradrenaline reuptake inhibitors (SNRIs), including Duloxetine. Caution is advised in patients taking anticoagulants and/or medicinal products known to affect platelet function (e.g., NSAIDs or acetylsalicylic acid (ASA)), and in patients with known bleeding tendencies.

Hyponatraemia: Hyponatraemia has been reported when administering Duloxetine, including cases with serum sodium lower than 110 mmol/l. Hyponatraemia may be due to a syndrome of inappropriate anti-diuretic hormone secretion (SIADH). The majority of cases of hyponatraemia were reported in the elderly, especially when coupled with a recent history of, or condition pre-disposing to, altered fluid balance. Caution is required in patients at increased risk for hyponatraemia, such as elderly, cirrhotic, or dehydrated patients, or patients treated with diuretics.

Discontinuation of Treatment: Withdrawal symptoms when treatment is discontinued are common, particularly if discontinuation is abrupt. In clinical trials, adverse events seen on abrupt treatment discontinuation occurred in approximately 45% of patients treated with Duloxetine and 23% of patients taking placebo.

The risk of withdrawal symptoms seen with SSRIs and SNRIs may be dependent on several factors, including the duration and dose of therapy and the rate of dose reduction. Generally, these symptoms are mild to moderate; however, in some patients they may be severe in intensity. They usually occur within the first few days of discontinuing treatment, but there have been very rare reports of such symptoms in patients who have inadvertently missed a dose. Generally, these symptoms are self-limiting and usually resolve within 2 weeks, though in some individuals they may be prolonged (2-3 months or more). It is therefore advised that Duloxetine should be gradually tapered when discontinuing treatment over a period of no less than 2 weeks, according to the patient's needs.

Elderly: Caution should be exercised when treating the elderly with the maximum dosage.

Akathisia/Psychomotor Restlessness: The use of Duloxetine has been associated with the development of akathisia, characterised by a subjectively unpleasant or distressing restlessness and need to move, often accompanied by an inability to sit or stand still. This is most likely to occur within the first few weeks of treatment. In patients who develop these symptoms, increasing the dose may be detrimental.

Concomitant use of other products containing duloxetine should be avoided.

Hepatitis/Increased Liver Enzymes: Cases of liver injury, including severe elevations of liver enzymes (>10-times upper limit of normal), hepatitis, and jaundice have been reported with Duloxetine. Most of them occurred during the first months of treatment. The pattern of liver damage was predominantly hepatocellular. Duloxetine should be used with caution in patients treated with other medicinal products associated with hepatic injury.

PREGNANCY AND LACTATION

Pregnancy: There are no adequate data on the use of Duloxetine in pregnant women. Studies in animals have shown reproductive toxicity at systemic exposure levels (AUC) of Duloxetine lower than the maximum clinical exposure.

The potential risk for humans is unknown. Epidemiological data have suggested that the use of SSRIs in pregnancy, particularly in late pregnancy, may increase the risk of persistent pulmonary hypertension in the newborn (PPHN). Although no studies have investigated the association of PPHN to SNRI treatment, this potential risk cannot be ruled out with Duloxetine, taking into account the related mechanism of action (inhibition of the re-uptake of serotonin).

As with other serotonergic medicinal products, discontinuation symptoms may occur in the neonate after maternal Duloxetine use near term. Discontinuation symptoms seen with Duloxetine may include hypotonia, tremor, jitteriness, feeding difficulty, respiratory distress and seizures. The majority of cases have occurred either at birth or within a few days of birth.

Duloxetine should be used in pregnancy only if the potential benefit justifies the potential risk to the foetus. Women should be advised to notify their physician if they become pregnant, or intend to become pregnant, during therapy.

Lactation: Duloxetine is very weakly excreted into human milk, based on a study of 6 lactating patients who did not breast-feed their children. The estimated daily infant dose on a mg/kg basis is approximately 0.14% of the maternal dose. As the safety of Duloxetine in infants is not known, the use of Duloxetine while breast-feeding is not recommended.

DRUG INTERACTIONS

Due to the risk of serotonin syndrome, Duloxetine should not be used in combination with non-selective, irreversible monoamine oxidase inhibitors (MAOIs) or within at least 14 days of discontinuing treatment with an MAOI. Based on the half-life of Duloxetine, at least 5 days should be allowed after stopping Duloxetine before starting an MAOI.